CHILLGUARD OPERATIONAL INTELLIGENCE

Technical Architecture, Codebase & System Documentation

System Version: 1.0 | Hackathon Prototype | Cold-Chain Logistics Monitoring

1. Executive Summary & System Overview

ChillGuard is an enterprise-grade operational intelligence dashboard engineered for real-time thermal monitoring, ML-driven breach forecasting, Mean Kinetic Temperature (MKT) audit compliance, and automated excursion prevention across cold-chain logistics networks (pharmaceuticals, food & produce, dairy, and seafood).

Unlike standard SaaS landing pages or generic admin portals, ChillGuard is built as a high-density operational platform following strict design and operational guidelines modeled after tools like Grafana, Datadog, and Linear.

2. High-Level Microservices Architecture

Subsystem Layer	Technology Stack	Core Responsibilities
Frontend App	React 18 + Vite 5 + TS Tailwind v3.4 + Framer Motion	Dashboard UI, Leaflet maps, Recharts graphs, Zustand state store, Socket.io client
Backend Server	Node.js 20 + Express 4 better-sqlite3 + Socket.io	REST API endpoints (/api/v1/*), SQLite WAL db, WebSocket broadcast, security middleware
ML Intelligence	Python 3.14 + FastAPI NumPy + Scikit-Learn + ReportLab	Breach risk classification (0-100), trajectory forecasting, Arrhenius MKT calculator, PDF report compiler
Telemetry Generator	Python 3.14 + Requests	Simulates 8 concurrent shipments with realistic physics & diurnal Karnataka temperatures

3. Design System & Hard Constraints

- **App Shell Background:** #F0F2F5 (Cool light gray, pure white layout background prohibited).
- **Sidebar:** #0D1B2A (Deep navy) with active item highlight #1D6FA4.
- **Typography:** DM Sans (Headings & Body) + IBM Plex Mono (Telemetry, IDs, Temperatures).
- **Status Colors:** Safe (#16A34A), At Risk (#D97706), Breach (#DC2626), Offline (#6B7280).
- **Zero Anti-Patterns:** No emojis, no radial background gradients, no spinning loaders (shimmer sweep loaders only), no bento grids.

4. Mathematical Physics & ML Intelligence Models

4.1 Breach Risk Classifier (ml_service/models/breach_classifier.py)

Evaluates 12 features in real-time to compute risk score $S \in [0, 100]$:

$S = \min(99, (S_{deviation} + S_{rate} + S_{door}) \times \gamma_{product})$

- $S_{deviation} = \min(45, |T_{current} - T_{setpoint}| / (T_{max} - T_{setpoint}) \times 40)$
- $S_{rate} = \Delta T_{15m} \times 120$ (for rising temp towards max)
- S_{door} = +25 if cargo door open
- $\gamma_{product}$ = 1.2 (Pharma), 1.1 (Seafood), 1.0 (Food/Dairy)

4.2 Arrhenius Mean Kinetic Temperature (MKT) Calculator

Calculates Arrhenius-weighted average kinetic temperature over transit history:

$T_{MKT} = (-\Delta H / R) / \ln(\Sigma \exp(-\Delta H / (R \times T_i)) / n) - 273.15$

Where ΔH = 83,144 J/mol (Pharma activation energy), R = 8.314 J/mol·K, T_i = Temp in Kelvin.

5. Pre-Seeded Demo Shipment Profiles

Shipment ID	Product	Route	Failure Event / Demo Purpose
SH-2041	Insulin (Biocon)	Bengaluru → Mysuru	Refrigeration fault at T+12m (Fires pre-excursion alert at Risk 78)
SH-2042	Alphonso Mangoes	Ramanagara → Chennai	Door open at T+8m (Shows door event marking & thermal rise)
SH-2043	Hepatitis B Vaccine	Bengaluru → Hubballi	No failure (Baseline normal thermal control)
SH-2044	Tilapia (Export)	Mangaluru → Bengaluru	Transit delay & slow rise (Gradual risk score increase)
SH-2045	Blood samples	Bengaluru → Kolar	Offline sensor at T+20m (Shows offline status handling)
SH-2046	mRNA Vaccine	Bengaluru → Delhi (Air)	Rapid breach (Shows critical breach excursion alert)
SH-2047	Dairy (Nandini)	Tumkur → Bengaluru	Completed shipment (Used for GDP compliance & PDF report demo)
SH-2048	Chemotherapy drugs	Mysuru → Bengaluru	Slow degradation (Predictive ML catches subtle thermal drift)

6. Local Execution Guide

1. **Python ML Service**: cd chillguard/ml_service && python main.py (Port 8000)
2. **Backend Express API**: cd chillguard/backend && npm run dev (Port 5000)
3. **Telemetry Simulator**: cd chillguard/generator && python simulator.py
4. **React Frontend Web App**: cd chillguard/frontend && npm run dev (Port 5173)